

Projective Constraints, Spectral Geometry, and Effective Dimensional Flow

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Abstract

The MPFS (Modèle Projectif Fractal Spectral) framework is an exploratory approach to the emergence of effective geometry from spectral principles. The present work investigates the interplay between projective constraints, spectral geometry, and dimensional flow within a mathematically structured spectral setting.

We first introduce a projective sector defined by a distinguished internal direction ($u = (1, 1, 1, -1, -1, -1)$) and an associated rank-one projector (Π). The effective spectral operator is constructed as $D_{\text{base}} = D_0 + \mu\Pi$, where (D_0) is a Dirac-type operator and (μ) denotes the projective coupling parameter. Using general results on finite-rank perturbations, we establish a Principal Spectral Theorem showing that the projective deformation preserves the Weyl asymptotics of the underlying geometry. Consequently, the ultraviolet spectral dimension remains equal to five and is independent of the projective coupling.

A geometric interpretation of this result is obtained through the orthogonal decomposition induced by the projective constraint. The distinguished direction generates a one-dimensional constrained sector, while the orthogonal complement defines an effective five-dimensional propagating space naturally associated with a compact manifold of toroidal type.

Since finite-rank projective perturbations cannot modify the leading spectral exponent, we introduce a fractal deformation of the spectral operator, $(D_F = |D_{\text{base}}|^{\alpha(\mu)})$, as a candidate mechanism for dimensional flow. Under this hypothesis, the spectral counting function acquires modified scaling properties, yielding an effective spectral dimension $(D_s = 5/\alpha(\mu))$. This relation provides a possible spectral mechanism for a continuous transition from a five-dimensional ultraviolet regime toward a lower-dimensional infrared regime.

The work further derives general consistency constraints on the fractal exponent, proposes a numerical program for testing the spectral predictions, and formulates explicit falsifiability criteria. While the projective sector is established on rigorous spectral grounds, the derivation of the fractal exponent from the fundamental projective-spectral axioms remains the central open problem of the MPFS framework.

Keywords: Spectral Geometry, Projective Constraints, Weyl Asymptotics, Spectral Dimension, Fractal Operators, Dimensional Flow, Dirac Operators, Mathematical Physics, MPFS Framework.

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1. Introduction

1.1 Motivation

One of the central questions of modern theoretical physics concerns the relationship between geometry, spectral structures, and the effective dimensionality of spacetime. While classical geometry characterizes space through local coordinates and metric properties, spectral geometry suggests that the fundamental information of a physical system may instead be encoded in the spectrum of an appropriate operator.

Several approaches to quantum gravity have indicated that the effective spectral dimension of spacetime may differ from its classical topological dimension. In particular, dimensional flow and scale-dependent geometry have emerged as recurring themes in causal dynamical triangulations, asymptotic safety, noncommutative geometry, and fractal models of spacetime.

These observations motivate the search for mechanisms capable of generating effective dimensions from an underlying spectral structure.

The MPFS (Modèle Projectif Fractal Spectral) framework explores this possibility by combining three fundamental ingredients:

1. A hierarchical spectral structure.
2. A projective internal sector.
3. A fractal deformation of the spectral operator.

The central objective is to investigate whether effective geometry and dimensional reduction can emerge from spectral principles alone.

1.2 Overview of the MPFS Framework

The starting point of the MPFS framework is the assumption of a spectral hierarchy

$$[\lambda_n = \lambda_0 q^n, \quad q > 1,]$$

which introduces a discrete scaling structure into the spectrum.

The projective sector is defined by the distinguished vector

$$u = (1, 1, 1, -1, -1, -1),$$

and the associated rank-one projector

$$\Pi = \frac{uu^\dagger}{u^\dagger u}.$$

The fundamental projective operator is

$$D_{\text{base}} = D_0 + \mu \Pi,$$

where (D_0) is a Dirac-type operator and (μ) is a projective coupling parameter.

The first part of this work investigates the spectral consequences of this projective deformation.

A second ingredient of the framework is the introduction of an effective fractal operator

$$\mathbf{D}_F = |D_{\text{base}}|^\alpha,$$

which provides a candidate mechanism for modifying the spectral dimension while preserving the underlying projective structure.

Within this setting, dimensional reduction is not imposed geometrically but is instead encoded in the spectral dynamics of the operator itself.

1.3 Objectives

The purpose of the present work is threefold.

First, we establish the principal spectral result of the projective sector. We prove that finite-rank projective perturbations preserve the Weyl asymptotics of the underlying geometry and therefore leave the ultraviolet spectral dimension unchanged.

Second, we provide a geometric interpretation of the resulting five-dimensional spectral behavior and show how an effective geometry naturally emerges from the projective structure.

Third, we investigate a fractal extension of the spectral operator and analyze its consequences for dimensional flow. In particular, we study the relation

$$\mathbf{D}_s = \frac{5}{\alpha},$$

which connects the spectral dimension to the fractal exponent.

The central open problem of the MPFS framework is the determination of the function

$$\alpha(\mu),$$

from the fundamental projective-spectral axioms.

The structure of the paper is as follows.

Section 2 establishes the Principal Spectral Theorem of the projective sector.

Section 3 derives the effective geometry associated with the projective constraint.

Section 4 introduces the fractal operator and the mechanism of spectral dimensional reduction.

Section 5 derives general constraints on the fractal exponent.

Section 6 presents a numerical program for testing the spectral predictions.

Section 7 discusses falsifiable predictions and possible observational consequences of the framework.

2. The Principal Spectral Theorem of the Projective Sector

Central Question

The projective sector of the MPFS framework is defined through the operator

$$D_{\text{base}} = D_0 + \mu\Pi,$$

where (D_0) is a Dirac-type operator and

$$\Pi = \frac{uu^\dagger}{u^\dagger u}$$

is the rank-one projector generated by

$$u = (1, 1, 1, -1, -1, -1).$$

The fundamental question addressed in this section is:

What is the ultraviolet spectral dimension of the projective sector?

More specifically, does the projective perturbation modify the asymptotic growth of the spectrum and therefore change the effective spectral dimension?

2.1 Geometric Framework

Let

$$T^5 = S^1 \times S^1 \times S^1 \times S^1 \times S^1$$

be the five-dimensional flat torus equipped with its standard Riemannian metric. We consider a self-adjoint elliptic Dirac-type operator

$$D_0: L^2(T^5, S) \rightarrow L^2(T^5, S)$$

acting on the spinor bundle S .

The classical Weyl theorem implies that the eigenvalue counting function

$$N_0(\Lambda) = \#\{\lambda_n : |\lambda_n| \leq \Lambda\}$$

Satisfies

$$N_0(\Lambda) \sim C \Lambda^5, \quad \Lambda \rightarrow \infty,$$

where $C > 0$ depends only on the geometry of T^5 .

Consequently,

$$D_s = \lim_{\Lambda \rightarrow \infty} \frac{\ln N_0(\Lambda)}{\ln \Lambda} = 5.$$

This value defines the ultraviolet spectral dimension of the reference sector.

2.2 Projective Structure

The projective sector introduces a distinguished internal direction

$$u = (1, 1, 1, -1, -1, -1).$$

The associated projector is

$$\Pi = \frac{uu^\dagger}{u^\dagger u}.$$

Since

$$u^\dagger u = 6,$$

the projector satisfies

$$\Pi^2 = \Pi, \quad \Pi^\dagger = \Pi.$$

Furthermore,

$$\text{rank}(\Pi) = 1.$$

The MPFS projective operator is then defined by

$$D_{\text{base}} = D_0 + \mu\Pi.$$

The parameter

μ

controls the strength of the projective deformation.

2.3 Spectral Stability Theorem

A central result of spectral theory states that bounded finite-rank perturbations do not alter the leading Weyl asymptotics.

Let

K

be a bounded operator of finite rank.

Then

$$N_{D_0+K}(\Lambda) = N_{D_0}(\Lambda) + O(1)$$

as

$$\Lambda \rightarrow \infty.$$

Consequently, the dominant power-law behavior of the spectrum remains unchanged.

Since

Π

is rank one,

$\mu\Pi$

belongs precisely to this class of perturbations.

2.4 Application to the MPFS Projective Sector

Applying the spectral stability theorem to

$$D_{\text{base}} = D_0 + \mu\Pi,$$

one obtains

$$N_{\text{base}}(\Lambda) = N_0(\Lambda) + O(1).$$

Using Weyl's law,

$$N_{\text{base}}(\Lambda) = C\Lambda^5 + O(1).$$

Therefore

$$\ln N_{\text{base}}(\Lambda) = \ln C + 5 \ln \Lambda + o(1).$$

The spectral dimension becomes

$$D_s = \lim_{\Lambda \rightarrow \infty} \frac{\ln N_{\text{base}}(\Lambda)}{\ln \Lambda} = 5.$$

The projective perturbation modifies only a finite number of spectral modes and cannot alter the asymptotic exponent.

2.5 Principal Spectral Theorem

Theorem.

Let

$D_{\text{base}} = D_0 + \mu \Pi$, where (D_0) is a Dirac-type operator on (T^5) and (Π) is the rank-one projector generated by

$$u = (1, 1, 1, -1, -1, -1).$$

Then

$$D_s^{UV} = 5$$

for all values of the projective coupling parameter μ .

2.6 Physical Interpretation

The theorem demonstrates that the projective sector alone does not produce dimensional reduction.

Although the projector modifies the spectrum locally, it cannot change the asymptotic growth law

$$N(\Lambda) \sim \Lambda^5$$

Therefore,

Projective constraints preserve the ultraviolet spectral dimension.

This result establishes the robustness of the five-dimensional ultraviolet regime within the MPFS framework.

2.7 Consequences for the MPFS Program

The Principal Spectral Theorem has two major consequences.

First, $D_s^{UV} = 5$ becomes a robust prediction of the projective sector.

Second, any transition of the form

$$5 \rightarrow 3$$

cannot originate from a finite-rank projective perturbation alone.

A new mechanism is therefore required.

This observation motivates the introduction of the fractal operator

$$D_F = |D_{\text{base}}|^\alpha,$$

which will be studied in the next sections as a candidate mechanism for spectral dimensional reduction.

3. Effective Geometry of the Projective Sector

Central Question

The Principal Spectral Theorem established that the ultraviolet spectral dimension of the projective sector is

$$D_s^{UV} = 5.$$

This result immediately raises a fundamental geometric question:

Why does the dimension 5 emerge in the MPFS framework?

The purpose of the present section is to show that the five-dimensional effective geometry arises naturally from the internal projective structure defined by the distinguished vector

$$u = (1, 1, 1, -1, -1, -1).$$

Rather than being postulated a priori, the effective dimension is shown to follow directly from the geometry of the projective constraint.

3.1 Internal Projective Space

The projective sector is formulated within an internal six-dimensional vector space $\mathcal{H}_{\text{int}} \simeq \mathbb{R}^6$.

The fundamental projective direction is represented by
$$u = (1, 1, 1, -1, -1, -1).$$

This vector defines a one-dimensional subspace
 $\langle u \rangle \lambda u; \lambda \in \mathbb{R}.$

The associated orthogonal projector is

$$\Pi = \frac{uu^\dagger}{u^\dagger u}.$$

Since (u) is nonzero, the projector selects a unique preferred direction inside the six-dimensional internal space.

3.2 Orthogonal Decomposition

The existence of the projective direction naturally induces the decomposition

$$\boxed{\mathbb{R}^6 = \langle u \rangle \oplus u^\perp.}$$

The orthogonal complement is defined by

$$u^\perp = \{x \in \mathbb{R}^6 : x \cdot u = 0\}.$$

Every vector

$$x \in \mathbb{R}^6$$

can be uniquely decomposed as

$$x = x_{\parallel} + x_{\perp},$$

with

$$x_{\parallel} \in \langle u \rangle,$$

and

$$x_{\perp} \in u^{\perp}.$$

Thus the projective sector naturally separates one distinguished direction from the remaining propagating degrees of freedom.

3.3 Effective Dimensional Reduction

Since

$$\dim(R^{\mathbb{G}}) = 6,$$

and

$$\dim(\langle u \rangle) = 1,$$

it follows immediately that

$$\boxed{\dim(u^{\perp}) = 5.}$$

This result is purely geometric and requires no additional dynamical assumptions.

The projective constraint effectively removes one direction from the propagating sector.

Consequently,

$$6 \rightarrow 5.$$

The number five therefore emerges directly from the projective decomposition itself.

3.4 Physical Interpretation

Within the MPFS framework, the direction

$$\langle u \rangle$$

plays the role of a constrained projective mode.

Fluctuations along this direction are controlled by the projector

$$\Pi.$$

The remaining physical degrees of freedom propagate in the orthogonal sector $u^{\perp}$.

Therefore, the effective geometry perceived by the spectrum is not six-dimensional but five-dimensional.

The dimensional reduction originates from the projective structure rather than from compactification or symmetry breaking.

3.5 Emergence of the Effective Manifold

The space

$$u^\perp$$

is a five-dimensional Euclidean vector space.

To obtain a compact spectral geometry, periodic identifications are introduced along each effective direction.

This corresponds to quotienting by a rank-five lattice $\Lambda \simeq \mathbb{Z}^5$.

The resulting compact manifold is

$$T^5 = u^\perp / \Lambda.$$

Equivalently,

$$T^5 = S^1 \times S^1 \times S^1 \times S^1 \times S^1.$$

The five-dimensional torus therefore appears as the minimal compact geometry naturally associated with the projective sector.

3.6 Spectral Consequences

The emergence of

$$T^5$$

provides a geometric explanation for the Weyl asymptotics obtained in Section 2.

Since the effective propagating space has dimension five, the spectral counting function satisfies

$$N(\Lambda) \sim C\Lambda^5.$$

Consequently,

$$D_s^{UV} = \lim_{\Lambda \rightarrow \infty} \frac{\ln N(\Lambda)}{\ln \Lambda} = 5.$$

The ultraviolet spectral dimension is therefore interpreted as the dimension of the effective propagating subspace selected by the projective constraint.

3.7 Geometric Principle of the MPFS Framework

The analysis of the projective sector leads to the following geometric principle:

A projective constraint removes one internal direction and generates an effective five-dimensional propagating geometry.

Within the MPFS framework, the sequence

$$u \implies \mathbb{R}^6 = \langle u \rangle \oplus u^\perp \implies \dim(u^\perp) = 5 \implies T^5 \implies D_s^{UV} = 5$$

provides the geometric foundation of the ultraviolet sector.

This interpretation establishes a direct connection between the projective structure and the spectral properties of the theory, thereby explaining the origin of the dimension five without introducing it as an independent assumption.

4. The Fractal Operator and Spectral Dimensional Reduction

Central Question

The Principal Spectral Theorem established that the projective operator

$$D_{\text{base}} = D_0 + \mu \Pi$$

preserves the Weyl asymptotics of the underlying geometry.

Consequently,

$$\boxed{D_s^{UV} = 5}$$

for all values of the projective coupling (μ).

This result raises a fundamental question:

If the projective sector preserves $D_s = 5$, how can dimensional reduction occur?

The answer is that finite-rank projective perturbations are mathematically incapable of modifying the leading spectral exponent. Any dimensional transition therefore requires a modification of the spectral scaling itself.

The purpose of this section is to introduce a candidate mechanism capable of generating such a transition.

4.1 Limitation of the Projective Sector

From Section 2, the spectral counting function satisfies

$$N_{\text{base}}(\Lambda) \sim C\Lambda^5.$$

The corresponding spectral dimension is

$$D_s^{UV} = 5.$$

The rank-one projector

$$\Pi$$

modifies only a finite number of spectral modes and cannot change the asymptotic growth law.

Therefore,

Projective constraints alone cannot generate dimensional reduction.

A new spectral mechanism is required.

4.2 Fractal Deformation Hypothesis

We postulate that the effective dynamics is governed not directly by

$$D_{\text{base}},$$

but by a fractal deformation of the spectrum.

The fundamental ansatz of the fractal sector is

$$D_F = |D_{\text{base}}|^{\alpha(\mu)}.$$

The exponent

$$\alpha(\mu) > 0$$

is interpreted as a fractal scaling parameter generated by the interaction between the projective sector and the spectral hierarchy.

The absolute value ensures positivity and guarantees that the fractional power is well-defined.

4.3 Spectral Transformation

Let

$$\lambda_n$$

denote the eigenvalues of

$$D_{\text{base}}.$$

The fractal operator transforms the spectrum according to

$$\lambda_n \longrightarrow \lambda_n^{(F)} = |\lambda_n|^\alpha.$$

Unlike the projective perturbation, which only shifts eigenvalues, the fractal deformation modifies the scaling law of the entire spectrum.

This is the key mechanism responsible for dimensional flow within the MPFS framework.

4.4 Spectral Counting Function

The counting function associated with the fractal operator is

$$N_F(E) = \#\{\lambda_n^{(F)} \leq E\}.$$

Using

$$|\lambda_n|^\alpha \leq E,$$

one obtains

$$|\lambda_n| \leq E^{1/\alpha}.$$

Therefore

$$N_F(E) = N_{\text{base}}\left(E^{1/\alpha}\right).$$

Applying the Weyl asymptotics of the projective sector,

$$N_{\text{base}}(\Lambda) \sim C\Lambda^5,$$

gives

$$N_F(E) \sim C\left(E^{1/\alpha}\right)^5.$$

Hence

$$\boxed{N_F(E) \sim C E^{5/\alpha}.$$

The exponent controlling the spectral growth has been modified.

4.5 Effective Spectral Dimension

The spectral dimension is defined by

$$D_s = \lim_{E \rightarrow \infty} \frac{\ln N_F(E)}{\ln E}.$$

Substituting the previous asymptotic expression yields

$$D_s = \lim_{E \rightarrow \infty} \frac{\ln C + \frac{5}{\alpha} \ln E + o(1)}{\ln E}.$$

Therefore

$$D_s(\mu) = \frac{5}{\alpha(\mu)}.$$

This relation constitutes the central result of the fractal sector.

Unlike the projective perturbation, the fractal deformation directly modifies the effective spectral dimension.

4.6 Ultraviolet Consistency

Consistency with the projective sector requires

$$\alpha(0) = 1.$$

In this limit,

$$D_F = D_{\text{base}},$$

and

$$D_s = \frac{5}{1} = 5.$$

Thus

$$\alpha = 1 \implies D_s = 5.$$

The ultraviolet fixed point of the theory is recovered.

4.7 Dimensional Reduction

Suppose that the fractal exponent evolves toward a larger value at lower energies.

A particularly significant case is

$$\alpha = \frac{5}{3}.$$

Using

$$D_s = \frac{5}{\alpha},$$

one obtains

$$D_s = \frac{5}{5/3} = 3.$$

Therefore

$$\boxed{\alpha = \frac{5}{3} \implies D_s = 3.}$$

The MPFS framework thus provides a simple spectral mechanism capable of generating a transition

$$\boxed{5 \rightarrow 3.}$$

The dimensional reduction emerges from spectral scaling rather than from geometric compactification.

4.8 Physical Interpretation

The MPFS framework separates two distinct physical mechanisms.

The projective sector determines the effective ultraviolet geometry and explains the appearance of the five-dimensional spectral regime.

The fractal sector modifies the scaling properties of the spectrum and generates a continuous flow of the effective dimension.

Schematically,

$$u \implies u^\perp \implies T^5 \implies D_s^{UV} = 5$$

for the projective sector, while

$$D_F = |D_{\text{base}}|^\alpha \implies D_s = \frac{5}{\alpha}$$

governs the dimensional flow.

This separation allows dimensional reduction to emerge dynamically without modifying the geometric foundation established in the previous sections.

4.9 Central Open Problem

The relation

$$D_s = \frac{5}{\alpha}$$

follows directly from the spectral scaling of the fractal operator.

However, the function

$$\alpha(\mu)$$

has not yet been derived from the fundamental MPFS axioms.

The principal unresolved question of the framework is therefore

Can $\alpha(\mu)$ be derived from the projective-spectral structure itself?

A successful derivation would transform the dimensional transition

$$5 \rightarrow 3$$

from a phenomenological hypothesis into a genuine prediction of the MPFS theory.

5. Constraints on the Fractal Exponent $(\alpha(\mu))$

Central Question

The fractal sector introduced in the previous section leads to the fundamental relation

$$D_s(\mu) = \frac{5}{\alpha(\mu)}.$$

The entire dimensional reduction mechanism of the MPFS framework therefore depends on a single quantity:

$$\boxed{\alpha(\mu)}.$$

This immediately raises the central theoretical question:

$$\boxed{\text{What determines } \alpha(\mu)?}$$

At present, the MPFS framework does not derive $(\alpha(\mu))$ from first principles. Nevertheless, the projective-spectral structure imposes a set of strong mathematical constraints that considerably restrict its possible form.

The purpose of this section is to establish these constraints and identify the minimal properties that any admissible function $(\alpha(\mu))$ must satisfy.

5.1 Ultraviolet Consistency Constraint

The projective operator

$$D_{\text{base}} = D_0 + \mu\Pi$$

preserves the Weyl asymptotics and therefore the ultraviolet spectral dimension

$$D_s^{UV} = 5.$$

The fractal deformation must reduce to the projective operator in the absence of fractal effects.

Therefore,

$$\mathbf{D}_F = D_{\text{base}}$$

When

$$\mu = 0.$$

This immediately implies

$$\boxed{\alpha(0) = 1.}$$

Substituting into

$$D_s = \frac{5}{\alpha}$$

gives

$$D_s(0) = 5.$$

The ultraviolet limit is therefore fixed uniquely.

5.2 Infrared Constraint

One of the primary motivations of the MPFS framework is the possibility of an effective three-dimensional infrared regime.

Assuming

$$D_s^{IR} = 3,$$

the spectral relation implies

$$3 = \frac{5}{\alpha_{IR}}.$$

Hence

$$\boxed{\alpha_{IR} = \frac{5}{3}.}$$

The infrared fixed point therefore determines the asymptotic value of the fractal exponent.

5.3 Monotonicity Requirement

The projective parameter

$$\mu$$

measures the strength of the projective deformation.

As the influence of the projective sector increases, the effective spectral dimension is expected to decrease continuously.

Since

$$D_s = \frac{5}{\alpha},$$

this requires

$$\alpha'(\mu) > 0.$$

Equivalently,

$$D_s'(\mu) < 0.$$

Thus the spectral flow must be monotonic:

$$5 \rightarrow 3.$$

without oscillations or discontinuities.

5.4 Boundedness

Combining the ultraviolet and infrared constraints yields

$$1 \leq \alpha(\mu) \leq \frac{5}{3}.$$

Consequently,

$$3 \leq D_s(\mu) \leq 5.$$

The theory predicts a bounded dimensional flow connecting two fixed points:

$$D_s^{UV} = 5, \quad D_s^{IR} = 3.$$

5.5 Scale Dependence

The projector

$$\Pi$$

introduces a single coupling parameter

$$\mu.$$

Dimensional analysis implies that (α) cannot depend directly on dimensional quantities but only on dimensionless ratios.

Introducing a characteristic projective scale

$$\Lambda_*,$$

one obtains

$$\alpha = \alpha\left(\frac{\mu}{\Lambda_*}\right).$$

This significantly restricts the possible functional dependence.

5.6 Minimal Analytic Model

The simplest smooth function satisfying all previous constraints is

$$\alpha(\mu) = 1 + \frac{2}{3} \left(1 - e^{-\mu/\Lambda_*}\right).$$

This function satisfies:

$$\alpha(0) = 1,$$

$$\alpha(\infty) = \frac{5}{3},$$

and

$$\alpha'(\mu) = \frac{2}{3\Lambda_*} e^{-\mu/\Lambda_*} > 0.$$

Substituting into the spectral dimension formula gives

$$D_s(\mu) = \frac{5}{1 + \frac{2}{3} (1 - e^{-\mu/\Lambda_*})}.$$

This provides a continuous interpolation between the ultraviolet and infrared regimes.

5.7 Geometric Interpretation

The exponent (α) can be interpreted as a measure of spectral fractality.

For

$$\alpha = 1,$$

the spectrum behaves like a conventional five-dimensional geometry.

For

$$\alpha > 1,$$

the eigenvalues grow more rapidly and the effective spectral dimension decreases.

The transition

$$\alpha: 1 \rightarrow \frac{5}{3}$$

therefore corresponds to a progressive reduction of the effective dimensionality of the spectral geometry.

5.8 Possible Origin of (α)

The constraints derived above do not explain why (α) should evolve.

A plausible interpretation is that (α) encodes the interaction between:

1. The projective sector,
2. The spectral hierarchy,
3. The fractal scaling structure.

Schematically,

$$(\Pi, q, \mu) \implies \alpha(\mu).$$

The explicit derivation of this relation remains unknown.

5.9 Main Open Problem

The previous constraints considerably restrict the possible behavior of the fractal exponent.

However, they do not yet provide a derivation.

The central unresolved problem of the MPFS framework can therefore be stated as follows:

Derive $\alpha(\mu)$ directly from the fundamental projective-spectral axioms.

A successful derivation would eliminate the only remaining phenomenological ingredient of the dimensional reduction mechanism and would transform

$$D_s = \frac{5}{\alpha(\mu)}$$

into a genuine prediction of the theory.

5.10 Summary

The fractal exponent is not arbitrary.

Consistency with the projective sector imposes:

$$\alpha(0) = 1,$$

$$\alpha(\infty) = \frac{5}{3},$$

$$\alpha'(\mu) > 0.$$

These conditions define a narrow class of admissible functions and establish the framework within which future derivations of $(\alpha(\mu))$ must be constructed.

The determination of $(\alpha(\mu))$ remains the central open problem of the MPFS program.

6. Numerical Spectral Analysis

Central Question

The analytical developments of the previous sections establish two central predictions of the MPFS framework:

$$D_s^{UV} = 5$$

for the projective sector, and

$$D_s(\mu) = \frac{5}{\alpha(\mu)}$$

for the fractal sector.

A natural question therefore arises:

Can these predictions be verified numerically?

The purpose of this section is to formulate a numerical program capable of testing the internal consistency of the MPFS framework and validating its principal spectral predictions.

6.1 Numerical Objectives

The numerical analysis has four primary objectives:

Objective I

Verify the Principal Spectral Theorem:

$$N(\Lambda) \sim C\Lambda^5.$$

Objective II

Confirm that finite-rank projective perturbations preserve the ultraviolet spectral dimension:

$$D_s^{UV} = 5.$$

Objective III

Test the fractal spectral law

$$N_F(E) \sim E^{5/\alpha}.$$

Objective IV

Investigate the dimensional flow

$$D_s: 5 \rightarrow 3.$$

predicted by the fractal operator.

6.2 Discretization of the Effective Geometry

The effective geometry derived in Section 3 is

$$T^5.$$

A convenient basis is provided by Fourier modes

$$k = (n_1, n_2, n_3, n_4, n_5), \quad n_i \in \mathbb{Z}.$$

The eigenvalues of the flat Dirac operator are approximated by

$$\lambda_k = \pm \sqrt{n_1^2 + n_2^2 + n_3^2 + n_4^2 + n_5^2}.$$

For numerical computations, the spectrum is truncated by imposing

$$|n_i| \leq N.$$

The resulting finite-dimensional Hilbert space contains

$$(2N + 1)^5$$

basis states.

Increasing (N) allows convergence toward the continuum limit.

6.3 Numerical Construction of the Projective Operator

The MPFS projective operator is

$$\mathbf{D}_{\text{base}} = D_0 + \mu\Pi.$$

The projector is

$$\Pi = \frac{uu^\dagger}{u^\dagger u},$$

with

$$u = (1, 1, 1, -1, -1, -1).$$

Since

$$\text{rank}(\Pi) = 1,$$

the perturbation modifies only a single spectral direction.

The full matrix representation of

$$D_{\text{base}}$$

is constructed numerically and diagonalized for various values of μ .

6.4 Spectral Counting Function

For each numerical spectrum

$$\{\lambda_n\},$$

the counting function is defined by

$$N(\Lambda) = \#\{|\lambda_n| \leq \Lambda\}.$$

The Weyl exponent is extracted from the logarithmic relation

$$\ln N(\Lambda) = A + D_s \ln \Lambda.$$

The effective spectral dimension is therefore estimated through

$$D_s^{\text{num}} = \frac{d \ln N}{d \ln \Lambda}.$$

A linear fit on a log-log plot directly yields the numerical value of (D_s) .

6.5 Verification of the Principal Spectral Theorem

The first numerical test consists of varying

$$\mu$$

while measuring

$$D_s^{\text{num}}.$$

The MPFS prediction is

$$D_s^{\text{num}} = 5$$

for all values of the projective coupling.

Although individual eigenvalues may shift, the asymptotic slope must remain unchanged.

Failure of this property would invalidate the Principal Spectral Theorem.

6.6 Numerical Implementation of the Fractal Operator

The fractal operator is defined by

$$D_F = |D_{\text{base}}|^\alpha.$$

After diagonalization of

$$D_{\text{base}},$$

each eigenvalue is transformed according to

$$\lambda_n \rightarrow |\lambda_n|^\alpha.$$

The resulting spectrum generates a new counting function

$$N_F(E).$$

The effective dimension is then measured from

$$\ln N_F(E)$$

as before.

6.7 Test of the Fractal Scaling Law

The central prediction of the fractal sector is

$$N_F(E) \sim E^{5/\alpha}.$$

Therefore

$$D_s^{\text{num}} = \frac{5}{\alpha}.$$

The numerical measurements can be directly compared with the analytical prediction.

Agreement would provide strong evidence supporting the fractal dimensional reduction mechanism.

6.8 Benchmark Cases

Three benchmark values are particularly important.

Ultraviolet Regime

$$\alpha = 1.$$

Prediction:

$$D_s = 5.$$

Intermediate Regime

$$\alpha = \frac{4}{3}.$$

Prediction:

$$D_s = 3.75.$$

Infrared Regime

$$\alpha = \frac{5}{3}.$$

Prediction:

$$D_s = 3.$$

These cases provide immediate consistency checks of the theory.

[6.9 Expected Numerical Outputs](#)

The numerical program should produce:

Spectral Density

$$\rho(\lambda).$$

Counting Function

$$N(\Lambda).$$

Log-Log Weyl Plots

$$\ln N \quad \text{vs} \quad \ln \Lambda.$$

Effective Spectral Dimension

$$D_s^{\text{num}}.$$

Dimensional Flow

$$D_s(\alpha).$$

These quantities provide a direct comparison between theoretical predictions and numerical observations.

[6.10 Numerical Falsifiability](#)

The numerical analysis offers a direct internal test of the MPFS framework.

The theory is falsified if one of the following occurs:

1. The projective sector fails to reproduce
$$D_s^{UV} = 5.$$

2. The fractal operator fails to produce
$$D_s = \frac{5}{\alpha}.$$

3. The dimensional flow deviates significantly from the predicted behavior.

Thus the numerical program is not merely illustrative; it provides a genuine consistency test of the theory.

6.11 Conclusion

The numerical analysis transforms the MPFS framework from a purely analytical construction into a computationally testable theory.

By directly measuring spectral dimensions and counting functions, numerical simulations can verify the central predictions

$$D_s^{UV} = 5$$

and

$$D_s = \frac{5}{\alpha}.$$

The numerical program therefore constitutes the first validation stage of the MPFS framework and provides the essential bridge between its mathematical structure and future phenomenological applications.

7. Falsifiable Predictions of the MPFS Framework

Central Question

A mathematically consistent framework is not necessarily a physical theory. To acquire predictive power, a model must generate consequences that can be confronted with numerical simulations, observational data, or future experiments.

This requirement naturally leads to the following question:

Can the MPFS framework be falsified?

The purpose of this section is to identify the principal predictions of the theory and to establish the conditions under which the MPFS framework would be confirmed or refuted.

7.1 Prediction I: Ultraviolet Spectral Dimension

The Principal Spectral Theorem predicts

$$D_s^{UV} = 5.$$

This result follows from the finite-rank nature of the projective perturbation

$$D_{\text{base}} = D_0 + \mu\Pi.$$

The corresponding spectral counting function satisfies

$$N(\Lambda) \sim C\Lambda^5.$$

Therefore, any numerical realization of the projective sector must reproduce the exponent

$$5.$$

Failure to recover this value would invalidate the geometric interpretation developed in Sections 2 and 3.

7.2 Prediction II: Fractal Spectral Scaling

The fractal sector predicts that the effective operator

$$\mathbf{D}_F = |D_{\text{base}}|^\alpha$$

changes the Weyl exponent according to

$$N_F(E) \sim E^{5/\alpha}.$$

Consequently,

$$\boxed{D_s = \frac{5}{\alpha}}.$$

This relation is one of the central quantitative predictions of the MPFS framework.

Numerical simulations must therefore recover the exponent

$$5/\alpha$$

within the expected numerical accuracy.

Any systematic deviation would falsify the proposed fractal mechanism.

7.3 Prediction III: Dimensional Flow

Combining the ultraviolet and infrared constraints yields

$$D_s^{UV} = 5, \quad D_s^{IR} = 3.$$

The MPFS framework therefore predicts a continuous dimensional flow

$$\boxed{5 \rightarrow 3}.$$

The flow is controlled by the fractal exponent

$$\alpha(\mu).$$

A successful numerical implementation should reproduce this transition as the effective projective coupling evolves.

7.4 Prediction IV: Characteristic Fractal Exponent

The constraints developed in Section 5 imply

$$\alpha(0) = 1,$$

$$\alpha(\infty) = \frac{5}{3}.$$

The value

$$\boxed{\alpha_{IR} = \frac{5}{3}}$$

plays a distinguished role because it corresponds to

$$D_s = 3.$$

Future derivations of $(\alpha(\mu))$ must remain compatible with these fixed points.

Any rigorous derivation leading to incompatible values would require a revision of the current MPFS framework.

7.5 Prediction V: Modified High-Energy Propagation

If the fractal spectral structure influences particle propagation, the effective dispersion relation may acquire corrections of the form

$$E^2 = p^2 \left[1 + \beta \left(\frac{p}{M_*} \right)^{\alpha-1} \right].$$

For the infrared value

$$\alpha = \frac{5}{3},$$

the correction becomes

$$\boxed{E^2 = p^2 \left[1 + \beta \left(\frac{p}{M_*} \right)^{2/3} \right].}$$

This prediction is not derived from the Principal Spectral Theorem itself but represents a phenomenological consequence of the fractal sector.

7.6 Time-of-Flight Signatures

The modified dispersion relation leads to an energy-dependent propagation velocity.

To leading order,

$$\Delta v \propto \left(\frac{E}{M_*} \right)^{2/3}.$$

Consequently,

$$\Delta t \propto L \left(\frac{E}{M_*} \right)^{2/3},$$

where (L) denotes the distance to the source.

Potential observational tests include:

4. Gamma-ray bursts,
5. Active galactic nuclei,
6. TeV gamma-ray sources,
7. Ultra-high-energy cosmic rays.

Future observations may place bounds on the scale (M_*).

7.7 Internal Consistency Tests

The MPFS framework contains several predictions that can be tested without reference to observational data.

These include:

$$N(\Lambda) \sim \Lambda^5,$$

$$N_F(E) \sim E^{5/\alpha},$$

$$D_s^{UV} = 5,$$

and

$$D_s = \frac{5}{\alpha}.$$

These relations provide a closed system of internal consistency checks.

7.8 Falsification Criteria

The MPFS framework would be falsified if any of the following statements were established:

Criterion A

Numerical simulations of the projective sector do not reproduce

$$D_s^{UV} = 5.$$

Criterion B

The fractal operator fails to generate

$$D_s = \frac{5}{\alpha}.$$

Criterion C

No admissible function

$$\alpha(\mu)$$

exists satisfying the ultraviolet and infrared constraints simultaneously.

Criterion D

The proposed fractal scaling is mathematically inconsistent with the spectral hierarchy underlying the MPFS framework.

Criterion E

Future high-energy observations exclude the predicted scaling behavior while simultaneously constraining all physically reasonable values of

$$M_{*}.$$

7.9 Predictive Status of the MPFS Framework

The predictions of the theory can be classified according to their degree of robustness.

Established within the Framework

$$D_s^{UV} = 5,$$

$$N(\Lambda) \sim \Lambda^5.$$

Derived within the Fractal Sector

$$D_s = \frac{5}{\alpha},$$

$$N_F(E) \sim E^{5/\alpha}.$$

Conditional Predictions

$$D_s^{IR} = 3,$$

$$\alpha_{IR} = \frac{5}{3}.$$

Phenomenological Consequences

Modified dispersion relations and time-of-flight effects.

7.10 Conclusion

The MPFS framework is not intended as a purely descriptive mathematical construction. Its central purpose is to generate quantitative spectral predictions that can be tested numerically and, potentially, observationally.

The theory therefore exposes itself to possible refutation through:

$$D_s^{UV}, \quad D_s(\alpha), \quad N(\Lambda), \quad N_F(E),$$

and the associated phenomenological consequences.

The existence of explicit falsification criteria is an essential feature of the MPFS program and provides a concrete pathway toward evaluating its physical relevance.

8. Conclusion and Outlook

The MPFS (Modèle Projectif Fractal Spectral) framework was developed as an exploratory approach to the relationship between spectral geometry, projective structures, and effective dimensionality.

The central result established in this work is the spectral stability of the projective sector. For the operator

$$\mathbf{D}_{\text{base}} = D_0 + \mu\Pi,$$

where

$$\Pi = \frac{uu^\dagger}{u^\dagger u}$$

is the rank-one projector generated by

$$u = (1, 1, 1, -1, -1, -1),$$

the Weyl asymptotics remain unchanged. Consequently, the ultraviolet spectral dimension satisfies

$$\boxed{D_s^{UV} = 5.}$$

This result is independent of the value of the projective coupling (μ) and follows from the general stability of Weyl asymptotics under finite-rank perturbations.

A geometric interpretation of this result was obtained through the decomposition $\mathbb{R}^6 = \langle u \rangle \oplus u^\perp$,

which naturally yields

$$\dim(u^\perp) = 5.$$

The effective manifold

$$T^5$$

therefore emerges as the minimal compact geometry associated with the propagating sector selected by the projective constraint.

Since finite-rank projective perturbations cannot modify the ultraviolet Weyl exponent, dimensional reduction requires an additional mechanism. This observation motivated the introduction of the fractal operator

$$\boldsymbol{D}_F = |D_{\text{base}}|^{\alpha(\mu)}.$$

Within this framework, the spectral counting function satisfies

$$N_F(E) \sim E^{5/\alpha},$$

leading to the effective spectral dimension

$$\boxed{\boldsymbol{D}_s(\mu) = \frac{5}{\alpha(\mu)}.$$

This relation provides a simple spectral mechanism through which a dimensional flow

$$5 \rightarrow 3$$

may emerge.

The analysis presented in this work also establishes several constraints on the fractal exponent, including

$$\alpha(0) = 1,$$

$$\alpha(\infty) = \frac{5}{3},$$

and

$$\alpha'(\mu) > 0.$$

These constraints significantly restrict the admissible class of fractal deformations and provide a framework for future investigations.

Numerical and phenomenological tests were proposed in order to evaluate the internal consistency of the theory and explore possible observational consequences. In particular, the MPFS framework predicts specific spectral scaling relations that can be investigated through numerical diagonalization of the effective operators.

Despite these developments, the framework remains incomplete. The principal unresolved problem is the determination of the fractal exponent from the fundamental projective-spectral structure itself.

Derive $\alpha(\mu)$ from the MPFS axioms.

This question represents the central challenge of the program.

A successful derivation would transform the dimensional flow

$$D_s = \frac{5}{\alpha(\mu)}$$

from a phenomenological hypothesis into a genuine prediction of the theory.

More broadly, the MPFS framework suggests that effective geometry may emerge from spectral organization rather than being imposed at the fundamental level. Whether projective spectral structures and fractal spectral dynamics ultimately provide a viable description of physical reality remains an open question, but the results presented here indicate that this possibility deserves further mathematical and numerical investigation.

Outlook

Several directions appear particularly promising for future research:

1. Derivation of the fractal exponent ($\alpha(\mu)$) from the projective-spectral axioms.
2. Rigorous spectral analysis of the operator

$$D_F = |D_{\text{base}}|^\alpha.$$

3. Large-scale numerical simulations of the spectral counting function.
4. Investigation of possible connections with dimensional flow scenarios in quantum gravity.
5. Development of quantitative phenomenological predictions capable of constraining the characteristic MPFS scale.

The resolution of these questions will determine whether the MPFS framework should be regarded as a useful spectral-geometric model or as a candidate physical theory with genuine predictive power.

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The MPFS framework developed in this work draws upon several established areas of mathematical physics, including spectral geometry, Weyl asymptotics, Dirac operators, spectral dimensions, and fractional spectral operators. The following references provide the mathematical and physical background relevant to the present study.

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